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The habitat requirements of hazel grouse (*Bonasa bonasia*) in managed boreal forest and applicability of forest stand descriptions as a tool to identify suitable patches

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Abstract

We evaluated preferences of habitat variables by hazel grouse (*Bonasa bonasia*) in an intensively managed boreal landscape in south-central Sweden. We surveyed 71 forest patches that varied in tree age, tree species composition, structure and field layer, as described by ordinary forest stand data. They were surveyed during autumn and spring for 2 years. A distinct difference in habitat quality, measured as amount of vertical cover and development of the field layer, was found among patches occupied during spring, patches occupied during only autumn and unoccupied patches. We found that the existing nation-wide forest stand data are adequate to select habitat patches to maintain hazel grouse in managed forested landscapes. However, the stand data would be improved if parameters concerning field-layer type, alder and field-layer cover were added. We used the results of an earlier study to predict hazel grouse occurrence in our studied patches. Unthinned middle-aged (20–69 years) or old (>90 years) stands rich in deciduous trees (5–40%), including alder, and with a rich field layer were the most often occupied patches. Moreover, the low density of hazel grouse in this landscape, compared to hazel grouse densities in less intensively managed landscapes, clearly showed the influence of forestry on the species.

Keywords: Habitat selection; Hazel grouse; *Bonasa bonasia*; Implications for management; Nation-wide forest stand descriptions

1. Introduction

Habitat structure has often been found to be one of the most important features for habitat selection by birds (Cody, 1981), and species conservation often focuses on preserving the preferred habitat. This approach relies on the assumption that individuals optimise their fitness by occupying a restricted part of the environment in their range (Wiens, 1989; Rosenzweig, 1991). Spatial scale is of great importance when studying a species' habitat preference (Rotenberry and Wiens, 1980; Åberg et al., 2000a). A multi-scale study of habitat use by hazel grouse (*Bonasa bonasia*) indicated that the home-range size was an appropriate scale to study this relationship, especially in fine-grained habitats (Åberg et al., 2000a).

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There are several difficulties in selecting relevant measures of habitat qualities (Wiens, 1995), and there are remarkably few studies on the precise consequences of present-day forestry on animals. For instance, the black woodpecker (*Dryocopus martius*) is considered to play a keystone role as a primary cavity excavator species, yet no studies have been performed to determine the effect of different forest practices on its occurrence (Mikusinski, 1995). This is surprising because forestry has been shown to strongly reduce the biodiversity within boreal forests in several studies (Berg et al., 1995; Esseén et al., 1997). Besides the need for information about species' responses to forestry, another important aspect for successful conservation of biodiversity is that the information must be expressed in a way that is understandable for forest managers (Noss, 1990; Angelstam, 1998). Consequently, existing nation-wide inventories of forests, known and applicable by forest officers, ecologists and landowners, would be of great value if they could be used to predict the occurrence of a species in the landscape and its response to different forestry practices.

The aim of this study was to document and define the habitat preferences of hazel grouse using Swedish forest stand descriptions in an intensively managed forested. We also tested the results of Swenson and Angelstam's (1993) regarding habitat criteria for hazel grouse and give specific applicable guidelines for maintaining the species in managed forested landscapes.

2. Materials and methods

2.1. Study area

The study area (Grimsö Wildlife Research Area) is 140 km^2 and located in south-central Sweden (59–60°N and 15–16°E), at the border of the boreal zone and the hemiboreal zone (Ahti et al., 1968). Forest dominates the area (72%), but bogs are also common (18%) (Cederlund, 1981). Forests have been intensively managed with thinning and selective removal of trees for several centuries (Wieslander, 1936). During the latter half of this century, clearcutting, planting of conifers and removal of deciduous trees transformed the forest stands into mostly even-aged, one-layered and single-tree-species monocultures. Norway spruce (*Picea abies*) and Scots pine (*Pinus sylvestris*) dominate the studied area. The proportion of deciduous trees, mostly white birch (*Betula pendula*), pubescent birch (*B. pubescens*), aspen (*Populus tremula*) and alder (*Alnus glutinosa* and *A. incana*) is less than 7% within these forests (Anonymous, 1999) and in stands >60 years old the deciduous component is usually <1% (Swenson and Angelstam, 1993). On the study area, the proportion of forest ages was approximately 29% young forest plantations or clearcut areas, 27% forest ready for thinning, 16% thinned forest, and 27% forest old enough to be clearcut (Anonymous, 1999). The field layer, herbaceous plants and half-shrubs with a maximum height of about 20 cm, was dominated by bilberry (*Vaccinium myrtillus*), cowberry (*V. vitis-idaea*), and wavy hairgrass (*Deschampsia flexuosa*).

2.2. Habitat description of the studied patches

Seventy-one patches, 15.5–40.0 ha in size (mean = 26.4 ha, S.D. = 5.3), were selected using forest stand description data accompanied by maps (1:10,000) supplied by the Swedish National Forest Enterprise. The patches consisted of one or several forest stands and were chosen to represent the variation in forest stands in the study area regarding tree age (except plantations), tree species composition, vegetation type and soil moistness. Areas that produce less than 1 m^3 wood per hectare annually were not considered as forest and are usually found on bogs and shallow soil over bedrock. The following habitat data were listed in the stand descriptions: the mean age and height of trees in each stand, the volume proportion of different tree species in the stand, the total standing volume, vegetation type (Anonymous, 1984), soil moistness, soil type and the proposed forestry measures to be conducted. The latter four habitat variables were listed as categories in the descriptions. The maps and descriptions were produced in 1986, using the standard methods in operational forestry in Sweden, described by The Swedish National Board of Forestry (Anonymous, 1988). For each of the 71 patches, additional fine-scale habitat data were collected during summer 1989. The sampling plots, a circle with a 10 m radius, were placed evenly at one per 2 ha in each patch. The variables measured or estimated in the plots were based on forest grouse studies by Marcström et al. (1983) and included: (1) the vertical cover estimated at four height intervals in the plot, 0–1, 1–4, 4–10 and >10 m. Also, the proportion of coniferous cover at the two lower heights was noted; (2) the density of the field-layer plant species known to be important in hazel grouse diet (Ivanter, 1962; Ahnlund and Selander, 1975; Eiberle and Koch, 1975; Wiesner et al., 1977) was estimated to the nearest 10%. These species were bilberry, cowberry, *Eriophorum* spp., *Viola* spp., *Anemone nemorosa*, *Oxalis acetosella* and *Potentilla erecta*. The latter four species were categorised into one group, herbs.

2.3. The hazel grouse

In Fennoscandia, the hazel grouse is dependent upon spruce-dominated multi-layered forests with a deciduous component, preferably alder (Swenson, 1993a; Swenson and Angelstam, 1993; Åberg et al., 2000a). The species is sedentary and territorial most of the year, shows low dispersal ability (Swenson, 1991a; Swenson and Danielsen, 1995) and is sensitive to habitat isolation (Åberg et al., 1995; Saari et al., 1998; Åberg et al., 2000b). Thus, the species shows several features, known and well described, which make it an appropriate species for studies of its habitat dependence (Beshkarev et al., 1994; Jansson and Angelstam, 1999). The hazel grouse is declining in Finland (Lindén and Rajala, 1981). As forestry in Finland has been similar to that in Sweden, it is important to access habitat conditions and the role of forestry on hazel grouse habitat in Sweden.

The 71 patches were surveyed for hazel grouse males in autumn 1988 and 1989, and in spring 1989 and 1990. No major forestry action (thinning, clear-cutting, etc.) was performed in these patches during the study period, although it may have occurred in adjacent stands. Surveying was conducted by walking through the areas in parallel lines 150 m apart, stopping at 150 m intervals, and imitating the territorial song of hazel grouse with a hunter's whistle for 6 min at a rate of two signals per minute. Each area was visited once per season between 15 September and 15 October in autumn, and 15 April and 15 May in spring, i.e. every patch was visited four times. The spring and autumn periods were chosen for surveying, because the hazel grouse are most territorial then, and respond best to the hunter's whistle (Swenson, 1991b). This method has a detectability of about 82%, based on tests on radio-marked birds and is described in more detail in Swenson (1991b).

2.4. Statistical analysis

The forest stand descriptions, together with the additional habitat variables measured in each patch, were used to analyse the habitat selection of hazel grouse males. The patches were divided into three quality categories, based on hazel grouse occurrence. Primary habitat was defined as patches with hazel grouse occurring during at least one spring breeding season, secondary habitat as patches having hazel grouse only during autumn(s) and non-habitat as patches where hazel grouse were not recorded. Possible differences in habitat variables among the three habitat categories were analysed in an ANOVA and tested with the Tukey HSD-test.

The results of Swenson and Angelstam (1993), which report that hazel grouse prefer habitats having a tree age of 20–69 years and a proportion of deciduous trees of 1–20% in managed forests, were examined both separately and combined (i.e. both the age and the deciduous tree species criteria) using the present data. The surveyed patches were grouped based on their properties, using the criteria reported by Swenson and Angelstam (1993) and the forest stand descriptions, into preferred and avoided patches, and the predicted and observed occurrence were compared. The study by Swenson and Angelstam (1993) was also conducted on the Grimsö Wildlife Research Area. A perfect fit to the predictions would be that all observed hazel grouse in the surveyed patches were detected in patches defined as preferred, that no patch defined as preferred was unoccupied, and that all others were unoccupied.

The density of hazel grouse males within the studied landscape was calculated using the forestry maps and the Swedish Yearbook of Forestry (Anonymous, 1999), in which the composition of the studied landscape concerning tree species composition and age is given. The proportion of suitable hazel grouse habitat in the study area was calculated using the criteria for suitable hazel grouse habitat based on this study.

3. Results

The number of patches occupied by hazel grouse was 17 in autumn 1988, 13 in spring 1989, 13 in autumn 1989, and 10 in spring 1990. The landscape consisted of approximately 7% suitable hazel grouse habitat. The total censused area was 1874 ha, constituting about 13% of the Grimsö Wildlife Research Area. The density of hazel grouse in the entire Research Area was approximately 0.36 males per 100 ha, using 20 ha as the mean size of a hazel grouse home range (Swenson, 1991c) and accounting for the differences in tree species composition and tree age between the studied patches and the studied landscape.

3.1. Differences among habitat categories

Significant differences in habitat variables were evident between primary habitat and non-habitat, but not between primary and secondary habitat, nor between secondary and non-habitat. Vertical cover at 4–10 m, total proportion of ground covered, proportion of spruce, and number of spruce per hectare were highest in the primary habitat. In contrast, the cowberry ground cover and proportion of pine were lowest in the primary habitat (Table 1). Also the proportion of deciduous trees was significantly higher in primary habitats than in non-habitats, when patches having greater than 50% deciduous trees were excluded. For all variables, the values for secondary habitats were intermediate between those for primary and non-habitat. Patches defined as primary habitat contained alder significantly more often than patches defined as non-habitat (χ^2-test $= 12.97$, $p = 0.00032$, d.f. $= 1$) and the type of field-layer cover was significantly richer in primary habitat than in non-habitat (Kruskal–Wallis test $H = 19.4$, $p = 0.022$, d.f. $= 9$).

3.2. Accuracy of prediction

The proportion of correctly predicted presence and absence of hazel grouse in the patches, using the tree age and deciduous component criteria of Swenson and Angelstam (1993) separately, was 73 and 55%, respectively (Table 2). None of the two single variable models over- or underestimated the presence of hazel grouse. We observed hazel grouse in 27 patches and the models predicted presence in 26 and 27 patches, respectively. Most of the wrongly predicted patches, using the tree age criteria, were due to hazel grouse being observed in older forest stands, mean of 94 years. Also, several of the wrongly predicted patches, using the criteria proportion of deciduous trees, were due to an absence of hazel grouse in patches with a low proportion of deciduous trees, mean 2.4%. When combining the two criteria, occurrence of hazel grouse

Table 1
Results of a one-way ANOVA, showing the habitat variables that were significantly different among hazel grouse primary, secondary and non-habitat patches[a]

Habitat variables	Habitat category		
	Primary, $n = 18$	Secondary, $n = 13$	Non-habitat, $n = 40$
From the forest inventory			
Proportion spruce	49.33*	40.92	27.85*
No. spruce per hectare	727.83*	439.46	394.85*
Proportion deciduous trees	13.56***	7.46	1.78***
Proportion pine	36.50**	51.62	65.70**
From the plot descriptions			
Vertical cover, 4–10 m	2.58**	2.39	1.91**
Proportion ground cover	1.80**	1.39	1.23**
Proportion cowberry	73.11*	87.62	91.40*

[a] Three patches were excluded when the effect of proportion of deciduous was analysed, see Section 3.
* Significant differences in F-values are signified as ($p < 0.05$).
** Significant differences in F-values are signified as ($p < 0.01$).
*** Significant differences in F-values are signified as ($p < 0.001$).

Table 2
Predicted and observed occurrence of hazel grouse in the 71 study patches, using predictions based on Swenson and Angelstam's (1993) study, regarding tree age, proportion of deciduous trees and combined criteria[a]

Observed values	Predicted values					
	Using age criteria		Using deciduous trees		Using combined criteria	
	Occupied, 20–69 years	Unoccupied, <20 and >70 years	Occupied, 1–20%	Unoccupied, <1 and >20%	Occupied	Unoccupied
Occupied	17	10	13	14	9	18
Unoccupied	9	35	14	30	7	37

[a] The proportion of correctly predicted habitat patches was 73, 60 and 65%, respectively.

was correctly predicted in 65% of the patches (Table 2). However, the last model underestimated the occurrence of hazel grouse, as it only predicted a total of 16 patches to be occupied by hazel grouse during the four seasons, in comparison to the 27 patches that were observed to be occupied.

4. Discussion

4.1. Comparison with earlier studies

Differences among the various habitat categories were found in the forest inventory variables showing that hazel grouse preferred patches with more spruce cover (proportion of spruce, density of spruce trees and proportion of pine (negative relationship)) and a greater proportion of deciduous trees. The plot descriptions also showed that the hazel grouse preferred patches with greater vertical cover (4–10 m), denser ground cover, and less cowberry cover. These results are in agreement with previous studies. Dense vertical cover was suggested to be important for hazel grouse survival from autumn to spring by Swenson (1991c), who found that the survival rate of hazel grouse during early winter was strongly and positively correlated with habitat density. This could be a key factor in the distinction between primary and secondary habitats. A possible explanation for the preference of the more dense areas is the avoidance of predators, for instance goshawks (*Accipiter gentilis*) (Lindén and Wikman, 1983). Similarly, the preference by the jay (*Garrulus glandarius*) (Andrén, 1990) and the black woodpecker (Rolstad et al., 1998) for stands having a high canopy closure was assumed to be related to the avoidance of goshawks. The amount of field-layer cover has been found to be a factor influencing hazel grouse presence in several earlier studies on hazel grouse (Seiskari, 1962; Eiberle and Koch, 1975; Marcström et al., 1983; Swenson, 1991c; Åberg et al., 2000a,b).

We interpret the results of the habitat analyses that the factors in Table 1, and presence of alder, are important clues for hazel grouse when choosing a territory. Our study confirms the results of Swenson and Angelstam (1993) that hazel grouse prefer habitats with intermediate proportions of deciduous trees, at least during the spring, and that the species is dependent on the presence of alder, as found by Swenson (1993a).

Almost two-thirds of the habitat patches were classified correctly regarding predicted and observed presence or absence of hazel grouse, using the two criteria from Swenson and Angelstam (1993). However, combining these criteria resulted in an underestimate, as only 16 patches were predicted to be occupied by hazel grouse during the four seasons, compared to the 27 patches that were observed to be occupied. Several of the occupied patches predicted to be empty had a deciduous proportion (21–36%) that was higher than the reported upper preferred limit, and some habitat was older than the preferred upper limit, i.e. stands >70 years (Swenson and Angelstam, 1993). However, these latter stands were considerably older, 86–110 years, and they may therefore have developed some multi-layered structure that attracted the hazel grouse. The mean proportion of deciduous trees in the patches predicted to be occupied, but which were observed to be empty, was very low, <5%, suggesting that this threshold is too low. Based on our study, we propose that the preferred hazel grouse habitats within managed forests should be 5–40% deciduous trees, preferably including alder, and of an age of 20–69 years or older than 90 years. Moreover, the stands should not have been thinned and the stands should have relatively rich vegetation with forbs and, preferably also *Vaccinium* species. These latter three variables are not explicitly given in the present stand descriptions, however.

There are at least four possible reasons for the differences in the reported preferred habitat for hazel grouse between this study and that of Swenson and Angelstam (1993). First, the methods differed. Swenson and Angelstam's (1993) 12 years of data were based on occasional observations of hazel grouse by personnel working in the study area. Also, the habitat description used in by Swenson and Angelstam (1993) was an average of that found in the 25 ha square where the hazel grouse were observed, not in the specific stand. In our study, an observation, i.e. a response of a male to the imitated song of another male, usually indicated that the area was at least worth defending (Swenson, 1991c) and the specific patch was described. Therefore our study probably gives a better indication of hazel grouse occurrence in given habitats, even though the earlier study in the area (Swenson and Angelstam, 1993) was based on a larger area. Secondly, Swenson and Angelstam (1993) only estimated cover indirectly using stand age, and their main aim was a comparison of use of the habitat features stand age and proportion of deciduous trees by three species of forest grouse. Thirdly, their habitat data were classified into categories, and wrongly classified habitat strongly can influence the result. Fourthly, the density of hazel grouse was higher during the study period of Swenson and Angelstam (1993) than during our study period (Åberg et al., 2000a), which according to theory of Fretwell and Lucas (1969), normally results in less specific habitat preferences. Therefore, we believe that our study more accurate describes the habitat preferences by the hazel grouse. Finally, our study was conducted on the scale of home-range size, which has been found to be a proper scale to study relationship between a species and its preferred habitat (Åberg et al., 2000a).

4.2. Conserving the most important habitats

The occurrence of hazel grouse can be predicted rather well using standard Swedish forest stand descriptions including cutting class (indicating cover), tree species composition and vegetation type, although presence of alder, whose presence in a stand not is given in the descriptions, is also important. However, the forest stand descriptions must be used with care, as their original purpose was mainly to provide information for forest production.

Our study suggests that the type of habitat preferred by hazel grouse during spring should receive the greatest attention by forest managers when creating and conserving suitable habitat. During this season, favourable hazel grouse habitat in managed boreal forest consists of non-thinned 20–69-year-old or old (>90 years) stands dominated by spruce and with about 5–40% deciduous trees. Also, the habitat patches, preferably larger than 25 ha, should especially contain richer field-layer cover and alder. Because the hazel grouse has been documented to be a poor disperser, these habitat patches should not be separated by non-suitable forest by more than 2 km in intensively managed landscapes or 200 m of open land (Åberg et al., 1995). Interestingly, studies on hazel grouse in Central Europe have showed that the species responds very quickly (10–20 years) to positive changes in their habitat (Scherzinger, 1979; Klaus, 1991), but unfortunately, its response to negative changes is perhaps just as quick (Swenson and Danielsen, 1991).

The response to intensity of forestry is obvious when comparing the density of hazel grouse in the intensively managed landscape we studied with the density in two comparable studies in less intensively managed boreal landscapes (Saari et al., 1998; Åberg, unpublished data). Our studied landscape, which was very intensively managed for pulp and timber, had approximately 0.4 males per 100 ha, when no consideration was given to the configuration of the habitat (i.e. habitat size or inter-patch distances), suggesting that the density in reality may have been even lower. In a less intensively managed forest owned mostly by small landowners near Sala, about 80 km north-east of this study area, a density of 0.7 males per 100 ha was found (Åberg, unpublished data). However, the density in both these studies were considerable lower than the density of 6 males per 100 ha found in a very lightly managed forest landscape in Finland (Saari et al., 1998).

In conclusion, this study suggests that (1) suitable hazel grouse habitat patches can be detected well using forest stand descriptions, especially if measurements of low vertical cover, presence of alder and type of field layer are added to the descriptions, (2) most conservation efforts should be put towards maintaining spring hazel grouse habitat, and (3) hazel grouse density can be used as a tool to measure the intensity of forestry and proportion suitable habitat left in a landscape. Moreover, Jansson and Andrén (1999) have shown that occurrence of hazel grouse may indicate a type of habitat also important for several other bird species, which further motivates conserving this type of habitat.

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